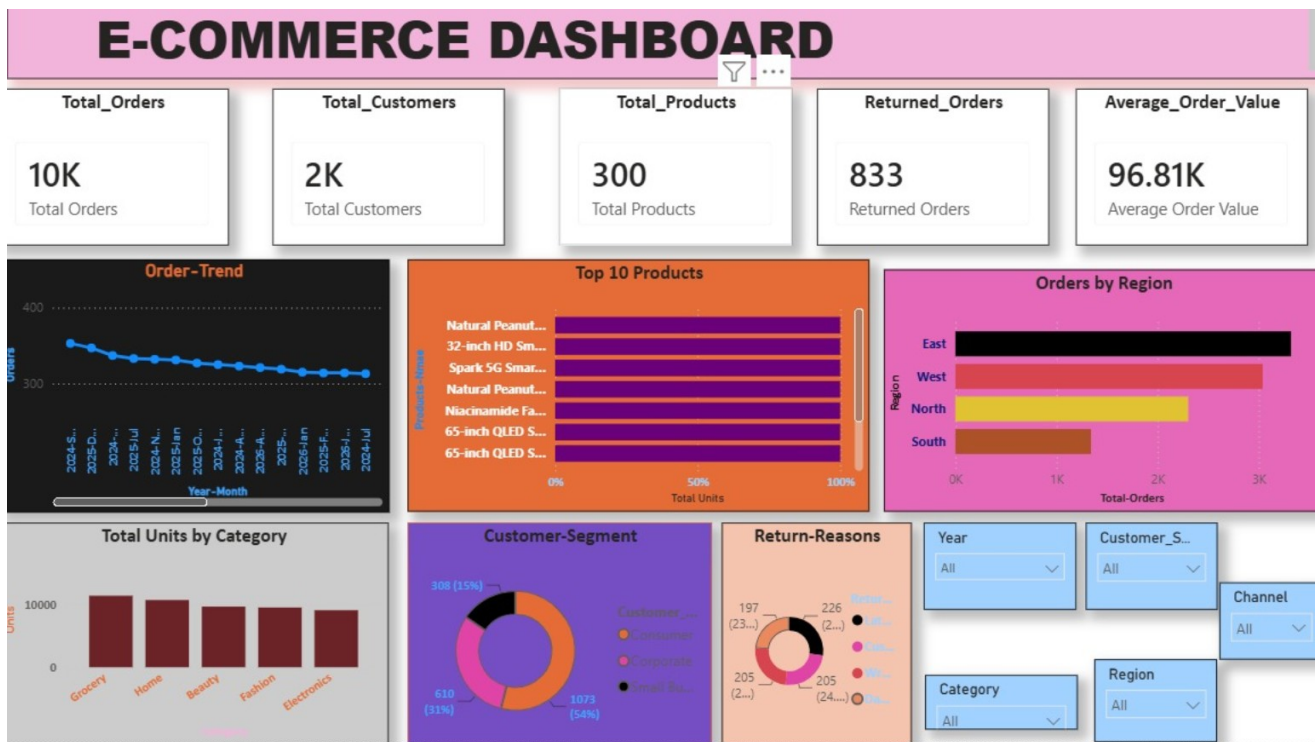


# E-COMMERCE BUSINESS 360

Power BI Dashboard – Project Documentation



**Tool:** Microsoft Power BI

**Project Type:** E-Commerce Data Analytics & Business Intelligence

**Prepared for:** Portfolio / GitHub

# 1. Project Overview

The E-Commerce Business 360 Dashboard is an interactive Power BI analytics project designed to provide a consolidated view of e-commerce order, customer, product, regional, category, and return-related performance. The dashboard uses KPI cards, trend analysis, product ranking, regional comparison, category analysis, customer segmentation, return-reason analysis, and interactive slicers.

# 2. Project Objectives

- Monitor overall order and customer activity.
- Track the number of products and returned orders.
- Understand average order value.
- Identify top-performing products by units.
- Compare orders across regions.
- Analyze units by product category.
- Understand customer-segment distribution.
- Analyze return reasons.
- Allow users to filter the dashboard by year, customer segment, channel, category, and region.

# 3. Dashboard KPI Snapshot

| KPI                 | Value shown in dashboard |
|---------------------|--------------------------|
| Total Orders        | 10K                      |
| Total Customers     | 2K                       |
| Total Products      | 300                      |
| Returned Orders     | 833                      |
| Average Order Value | 96.81K                   |

These values are transcribed from the supplied dashboard screenshot and are presented as the dashboard's displayed KPI values.

## 4. Dashboard Components

| Component               | Purpose                                                                     |
|-------------------------|-----------------------------------------------------------------------------|
| Total Orders            | Shows the overall number of orders represented in the dashboard.            |
| Total Customers         | Shows the total customer count represented in the dashboard.                |
| Total Products          | Shows the number of products represented.                                   |
| Returned Orders         | Shows the number of returned orders.                                        |
| Average Order Value     | Displays the average order value KPI.                                       |
| Order Trend             | Shows order movement across year-month periods.                             |
| Top 10 Products         | Highlights products ranked by total units.                                  |
| Orders by Region        | Compares total orders across East, West, North, and South.                  |
| Total Units by Category | Compares units across Grocery, Home, Beauty, Fashion, and Electronics.      |
| Customer Segment        | Shows the distribution of Consumer, Corporate, and Small Business segments. |
| Return Reasons          | Shows the distribution of return reasons.                                   |
| Slicers                 | Enables filtering by Year, Customer Segment, Channel, Category, and Region. |

## 5. Data Preparation & Power BI Workflow

**Step 1 – Data Import:** The source data is imported into Power BI for analysis.

**Step 2 – Data Cleaning:** Data types, missing values, duplicate records, and inconsistent values should be checked in Power Query before modeling.

**Step 3 – Transformation:** Required columns and analytical fields are prepared using Power Query transformations.

**Step 4 – Data Modeling:** Tables and relationships are organized so that order, customer, product, category, region, and return analysis can be performed correctly.

**Step 5 – DAX Measures:** Measures are created for dashboard KPIs such as Total Orders, Total Customers, Total Products, Returned Orders, Average Order Value, and Total Units.

**Step 6 – Visualization:** KPI cards, line charts, bar charts, donut charts, and slicers are arranged into a one-page interactive dashboard.

## 6. Tools & Technologies

- Microsoft Power BI
- Power Query
- DAX (Data Analysis Expressions)
- Microsoft Excel / tabular source data
- Data Visualization & Business Intelligence

## 7. Interactive Filtering

The dashboard includes slicers for Year, Customer Segment, Channel, Category, and Region. Users can select values to dynamically update the dashboard visuals and KPIs.

## 8. Dashboard Observations

Based on the supplied dashboard screenshot, the dashboard displays 10K total orders, 2K total customers, 300 total products, 833 returned orders, and an average order value of 96.81K.

The regional chart provides a direct comparison of order volume across East, West, North, and South. The category chart compares total units across Grocery, Home, Beauty, Fashion, and Electronics.

The customer-segment donut chart separates customers into Consumer, Corporate, and Small Business segments, while the return-reasons chart provides a breakdown of reasons associated with returned orders.

The Top 10 Products visual is intended to identify products with high total-unit performance, while the Order Trend visual provides a time-based view of order activity.

## 9. Business Use Cases

- Management can monitor high-level e-commerce KPIs from a single page.
- Business teams can compare regional order performance.
- Product teams can identify products with high unit volume.
- Customer teams can understand customer-segment distribution.
- Operations teams can review returned orders and return reasons.
- Users can apply slicers to investigate specific years, regions, categories, channels, or customer segments.

## 10. Conclusion

This project demonstrates the use of Power BI for transforming e-commerce data into an interactive Business 360 dashboard. It combines KPI reporting, trend analysis, product analysis, regional analysis, category analysis, customer segmentation, return analysis, and interactive filtering in a single-page report.

## 11. Portfolio Files

- Power BI .pbix dashboard file
- Source dataset (where permitted)
- Dashboard screenshot
- This project documentation PDF
- GitHub README.md